

Digital Therapeutics Evolution What kind of Research Will Make the Difference in this Area?

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ABSTRACT

As in prescribable medical drug-based therapies, Digital Therapeutics (DTx) solutions introduce the use of software as 1) an active ingredient implemented as digital interventions to improve patients' condition; and 2) as the excipient through which the intervention is conveyed to the patient. The most common DTx solutions implemented until today deal with the use of software-based solutions for dealing with mental health conditions. However, the potential of DTx is envisioned to grow to treat other health conditions with a number of technologies that still have not been duly explored. The objective of this workshop is to discuss the opportunities, barriers, and challenges of future DTx, the envisioned future use-cases, and the technologies with high potential to drive the uptake of these technologies, and this way to position this field as a solid complement of today's therapeutic approaches for dealing with different health conditions.

CCS CONCEPTS

• **Interaction techniques**; • **ubiquitous and mobile devices**; • **design and evaluation methods**;

KEYWORDS

Digital health, pervasive health systems, artificial intelligence for health

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1 RATIONALE OF THE WORKSHOP OBJECTIVES

The latest advances in technology and regulatory frameworks on healthcare enable the possibility to shift from the concept of digital medicine, in which digital technologies are used as a tool for supporting therapy (i.e., telemonitoring), to the recently developed concept of Digital Therapeutics (DTx). In DTx the medical treatment is based on the use of software and data, or software- and data-driven solutions, in conjunction with hardware devices and traditional medical therapies such as drugs [1] [2]. The new era of DTx [3] can find an important tool in artificial intelligence through which a more personalized form of healthcare can emerge thanks to the capability to dynamically adapt to patients' individual clinical needs, goals, and lifestyles [4]. However, to achieve the full potential of DTx solutions there are still several open challenges to address.

Health is one of the main applications areas of ubiquitous technology with a huge potential impact on the society. Medicine can sometimes be conservative, and while consumer devices and applications have long embraced this technology to drive applications for health management and behavior change, DTx solutions – which can be considered their clinical counterpart – are still in their infancy. With a focus on the development and design of new DTx solutions, the present workshop aims to contribute to the dissemination and discussion of pervasive and ubiquitous computing technologies. In addition, by gathering insights and providing a space for knowledge sharing between researchers, students, and other enthusiasts of DTx, we aim to contribute to the identification of practical, large-scale solutions that can be easily implemented in real-world settings.

1.1 Workshop Objectives

The objective of this workshop is to address the main challenges for achieving the wider potential of future DTx, and their potential impacts to healthcare management by discussing:

- Which are the main design challenges for future DTx solutions?
- Which are the best enabling technologies (i.e., the digital excipients) to improve the outreach of DTx?
- Which characteristics of digital interventions (i.e., digital active ingredients) are the key components for achieving clinical efficacy, and for what kind of clinical conditions?
- What should the role of ubiquitous technology be in DTx?

- How can knowledge on AI, HCI and clinical sciences be combined to design future DTx solutions?
- Which can be (and why?) the best use-cases beyond the current ones (i.e. mental health) for the wider uptake of DTx?
- How could these technologies be incorporated into existing clinical practice processes to maximise their uptake?
- What are the key sustainability aspects to consider for future DTx implementation?
- How much data is important in DTx? Furthermore, how to address the bias and trustworthiness in the AI algorithms?

2 WORKSHOP ORGANISERS

Oscar Mayora Ibarra: Prof. Mayora is head of Digital Health Research Unit at Fondazione Bruno Kessler (FBK) in Trento, Italy. He graduated in Electronics and Telecommunications and obtained a master's degree in computer science at Tecnológico de Monterrey and received a Ph.D. in Electronic Engineering and Informatics at DIBE, University of Genoa, Italy in 2000. In the same year he was awarded an ERCIM Visiting Research Fellowship to conduct post-doctoral activities at the Advanced Interactive Systems Laboratory of VTT Electronics in Oulu, Finland. In August 2001 he was appointed Associate Professor in Computer Science at Tecnológico de Monterrey where he became head of the Graduate Program in Computer Science. Later from 2004 – 2016 he was Head of the Ubiquitous Technologies for Health Area at Create-Net. During the past years he has conducted research activity in the field of Ubiquitous and Pervasive Health Technologies. Since 2005, Dr. Mayora has been actively participating in international research projects in the digital health domain where he has coordinated over 7 projects funded by the European Commission and participated in more than 10. Dr. Mayora is scientific coordinator for the Joint Research Unit between the Trentino Local Health Trust, the Province of Trento and FBK as part of Trentino Salute 4.0 Initiative.

Bert Arnrich: Prof. Dr.-Ing. Bert Arnrich is head of the Chair Digital Health - Connected Healthcare at the joint Digital-Engineering Faculty of Hasso Plattner Institute (HPI) and the University of Potsdam. The research group Digital Health - Connected Healthcare aims to pave the way for transforming healthcare systems from purely managing illness to maintaining wellness. Ubiquitous sensing and computing technologies are foreseen as the key enabler for pushing the paradigm shift from the established provider-centric healthcare model to a user-centered and preventive overall lifestyle health management that is available everywhere, anytime and to anyone. Bert Arnrich has been a PI in several European and national research projects. He studied "Informatics in the Natural Sciences". In his PhD thesis he implemented an early big data approach that collects and consolidates patient data for scientific data analysis. At ETH Zurich he established and headed the research group Pervasive Healthcare in the Wearable Computing Laboratory. He received a Marie Curie Cofund Fellowship from the European Union and was appointed to tenure track professorship at the Computer Engineering Department at Bosphorus University. He worked as a Science Manager for Emerging Technologies at Accenture Technology Solutions.

Hristijan Gjoreski: Associate Prof. at the Ss. Cyril and Methodius University in Skopje, N. Macedonia, and is lead data scientist at

Emteq Labs, UK. He finished his PhD in 2015 at the Jozef Stefan Institute in Slovenia and was postdoctoral researcher in 2017 at University of Sussex, UK. His scientific and research experience is in the domains of Machine Learning and applied Artificial Intelligence. He has specialized in development of machine-learning algorithms in the areas of wearable computing, e-health, activity recognition and affective computing. He has received the award "Best Young Scientist" for 2016 from the President of Republic of Macedonia. Hristijan was featured in the list of Top 2% scientists in the world in the category single-year impact award. Hristijan and his teams have won 3 international machine learning competitions for human activity recognition with wearable sensors. He has participated in more than 12 international projects, and he is currently the coordinator of the WideHealth European H2020 TWINNING project (no. 952279). He established and is organizer of the Data Science Macedonia group, with more than 1100 members.

Tiago Guerreiro: Associate Professor at Universidade de Lisboa (Faculdade de Ciências) and a researcher and Vice-Director at LASIGE. His main areas of expertise are Pervasive Healthcare, Mobile and Accessible Computing. In these areas, he has published 100+ peer-reviewed papers. He received awards for 10+ papers and an ACM Best of Computing award in 2016. He is Editor-in-Chief for ACM Transactions on Accessible Computing and was ASSETS 2020 General Chair, among many other service roles. He participated in 10+ EU and 1 IMI project. He was an expert evaluator for H2020 EU, EIT Health, CMU Portugal, and ERA ICPermed calls and was an invited expert supporting the European Commission in implementing the Web Accessibility Directive. He is the president of his institution's Ethics Review Board and the coordinator of the master's in data science.

Filipa Ferreira-Brito: Neuropsychologist and a member of the Portuguese Association of Psychologists. She holds a Bachelor's degree in Psychology from the University of Évora and a Master's in Neuropsychology from the Catholic University of Lisbon, Portugal. She is finishing her PhD in the EnviHealth & Co Doctoral Program at the Faculdade de Medicina da Universidade de Lisboa, Portugal, while working as a research assistant at LASIGE, Department of Informatics, Faculdade de Ciências da Universidade de Lisboa, Portugal. Her research work focuses on designing digital health platforms to assess and promote health-related outcomes, which has resulted in several international journal publications as both author and co-author. She has been recognized for her work through the ISAMB/AstraZeneca scientific incentive award in 2019 and two LASIGE Distinguished Papers Awards for works published as first author in 2019 and 2020.

Mitja Luštrek: Senior Research Associate and the Head of the Ambient Intelligence Group at the Department of Intelligent Systems at Jožef Stefan Institute, Ljubljana, Slovenia. He holds his PhD degree from the Faculty of Computer and Information Science of the University of Ljubljana. His main research interests are the interpretation of sensor and other health-related data using machine learning, and development of personal health systems and DTx using sensing and artificial intelligence. He has been the principal investigator in numerous international research projects on these topics. He teaches ambient intelligence at Jožef Stefan International Postgraduate School. He has been successful at many scientific competitions, such as twice winning the Sussex-Huawei

Locomotion-Transportation Recognition Challenge and being one of the two grand winners of the \$500k XPrize Pandemic Response Challenge

3 EXPECTED ATTENDANCE AND PLAN FOR THE WORKSHOP

The workshop would expect to accept a maximum of 10 outstanding contributions discussing the key challenges, opportunities, and strategies for the implementation of future DTx solutions. In addition, the workshop will discuss the most promising technologies to be used as digital excipients (i.e. AI-empowered chatbots, virtual reality, etc.) and the best candidates of clinical interventions to be used as active ingredients (i.e. Cognitive Behavioral Therapies, Behavior Activation, Nutrition guidelines, prevention strategies, etc.).

The workshop submissions will be evaluated by a panel of a minimum of two reviewers coordinated by one of the workshop co-authors. The workshop will be organized in a full day onsite modality in which the 10 highest evaluated submissions will be invited to present their contributions during the first half of the day. The second part of the workshop will consist in a guided discussion

around the key questions identified in the objectives regarding the uptake of future digital therapeutics technologies. It is envisioned the production of a final report with the workshop outcomes that will be evaluated to be submitted for further publication, such as a contributing note to a relevant journal or as the basis for a journal special issue.

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